THE NORMALIZATION IS IDEMPOTENT: NORMALIZING THE NORMALIZATION GIVES THE SAME SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-normalization-is-normalization is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 14050-14055 and proof span 14057-14064 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof ends with the author's sentence 'We omit the verification that the morphisms are mutually inverse'; that sentence is kept verbatim and what it leaves to the reader is the routine use of uniqueness of the factorization, the mechanism of which is stated in the counted span.

Editorial note. 1 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The characterization of the normalization as the universal object for factorizations through a locally birational integral morphism, cited from the same chapter. Cited at proof lines 14057-14061. Applies to the composite of the second normalization with the first to produce a canonical morphism between them.
- The uniqueness part of the same characterization, which is what makes the two morphisms mutually inverse. Cited at proof lines 14062-14064. Identifies the canonical morphism and its companion as inverse to one another.

Source reference. Original lemma 035I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4129–4129. Same-chapter reference used in the retained author text at source line(s) 14027, 14033, 14059, 14409, 14538, 14556, 14913, 15004. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $f : Y \to X$ be a quasi-compact and quasi-separated morphism of schemes. Let X' be the normalization of X in Y. Then the normalization of X' in Y is X'.*

Proof. If $Y \to X'' \to X'$ is the normalization of X' in Y, then we can apply Lemma 035I to the composition $X'' \to X$ to get a canonical morphism $h : X' \to X''$ over X. We omit the verification that the morphisms h and $X'' \to X'$ are mutually inverse (using uniqueness of the factorization in the lemma). □